



POLICY BRIEF

CLEAN ENERGY & POWER SERIES: G3

ADDRESSING ERRATIC AND LOW-PRESSURE WIND FLOW IN WIND POWER PLANTS

Md Hasin Israq | Hasibul Hasan Ahmed

Mst. Fatema Binta Alam Jim | Md Samiur Rahman Sazzad

Key Findings

1. Bangladesh's coastal regions have high wind energy potential, but progress is limited by high costs, infrastructure gaps, and reliance on imported technology.
2. Combating the erratic nature of renewable sources is essential to ensure reliable energy supply, build public confidence, and increase adoption.
3. VAWTs and hybrid wind-solar systems are well-suited to Bangladesh's low and variable wind conditions, with centralized storage enhancing grid stability.

Key Policy Recommendations

1. Invest in VAWTs and hybrid wind-solar projects for resilient, continuous energy supply.
2. Establish centralized storage and shared infrastructure to stabilize supply and reduce costs.
3. Promote local R&D, public awareness, and community involvement to foster support and reduce reliance on imports.

Introduction

Bangladesh is at a critical point in its energy transition, motivated by the necessity to diversify its energy portfolio, diminish dependence on imported fossil fuels, and strengthen resilience to climatic threats. The nation's existing energy infrastructure is predominantly dependent on fossil fuels, which are expensive and unsustainable amid increasing demand. The pressing demand for robust, sustainable solutions positions wind energy, particularly in the coastal and offshore areas of Bangladesh, as a central focus. The Bay of Bengal region, exhibiting wind speeds ranging from 5.75 to 7.75 m/s, possesses significant, but mostly unexploited, potential for sustainable energy generation (IEPMP, 2023). Nonetheless, obstacles such as elevated expenses, insufficient resilient infrastructure, and restricted public awareness persist in impeding advancement. This policy brief outlines a collection of unique, locally-adapted solutions that tackle these issues. Our initial plans emphasise the implementation of advanced hybrid technologies, the promotion of community engagement, and the establishment of economic support mechanisms to position wind energy as a fundamental component of Bangladesh's sustainable energy portfolio, thereby enhancing energy security and resilience for all.

Current Scenario

Wind energy in Bangladesh has not yet substantially influenced the nation's energy framework. Bangladesh has a mere 63 MW of installed wind energy capacity, predominantly from a single large-scale project, indicating a deficiency in wind energy adoption (Joshi et al., n.d.). Projects like the 55 MW Mongla wind initiative have proven the technological and commercial viability of wind energy, producing electricity at competitive prices (The Daily Star, 2022). However, isolated and rural regions, especially coastal ones, continue to depend on unreliable fossil-fuel-based electricity supplies.

The use of wind energy, bolstered by sophisticated hybrid systems and community-oriented policies, has the potential to transform Bangladesh's energy industry. Our solutions seek to use the nation's wind potential through the implementation of contemporary, cyclone-resistant technologies and the augmentation of grid resilience to guarantee dependable energy access for all communities.

Challenges & Barriers

1. Uneven Distribution and Limited Transmission Infrastructure

Wind energy resources in Bangladesh are concentrated along coastal regions, far from high-demand cities such as Dhaka. Offshore wind farms in these places require a strong transmission network to efficiently transfer energy to the national grid, where grid infrastructure is inadequate (Chowdhury, 2020). Furthermore, frequent natural disasters expose isolated areas to energy outages, as current system resilience is insufficient to survive such occurrences.

2. High Initial Costs and Dependency on Imported Technology

The capital-intensive nature of wind energy projects, along with reliance on imported technology, creates a considerable obstacle. This reliance on foreign equipment and expertise raises costs, constrains local job development, and reduces scalability. Traditional horizontal-axis wind turbines (HAWTs), which are widely used in early projects, are frequently ineffective in low and variable wind conditions, rendering them unsuitable investments in Bangladesh's particular environment (Miskat et al., 2020).

3. Public Perception and Community Engagement

Wind energy initiatives can only be successful with the backing of the public. But locals are against it because of worries about land use, environmental effects, and dependability. Coastal locations are especially vulnerable to natural disasters, and because of their lack of exposure to contemporary, resilient technologies, rural and remote communities are understandably sceptical about the reliability of wind energy. Acquiring public support for wind energy requires raising awareness and encouraging participation in the benefits of the technology (Chowdhury, 2020).

4. Insufficient Energy Storage and Grid Modernization

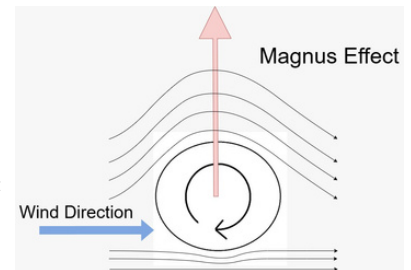
The inadequate energy storage infrastructure limits the gradual integration of renewable sources such as wind and solar into Bangladesh's grid. The variable characteristics of wind energy necessitate advanced storage devices to ensure a consistent power supply, especially in isolated regions vulnerable to grid disruptions during severe weather (IEPMP, 2023). Enhancing grid resilience and storage is crucial for stabilising energy supply as renewable sources gain prominence.

Scientific Solution

The approach we offer presents innovative, context-specific technologies and hybrid systems designed for Bangladesh's unique environmental circumstances, tackling unpredictable wind patterns and improving grid resilience.

1. Vertical Axis Wind Turbines (VAWTs)

Vertical Axis Wind Turbines (VAWTs) provide a solution specifically tailored to the low and erratic wind conditions of Bangladesh. In contrast to Horizontal Axis Wind Turbines (HAWTs), Vertical Axis Wind Turbines (VAWTs) may function effectively at low wind speeds and variable directions, proving them suitable for the coastal regions of Bangladesh. The Magnus VAWT utilizes the Robins-Magnus Effect, demonstrating efficacy in both low wind velocities and high-pressure scenarios, hence maintaining resilience during cyclonic occurrences. The Magnus VAWT has been successfully installed in the Philippines, demonstrating its efficacy in comparable hard wind conditions (Babu et al., 2022).



Vertical Axis Wind Turbines (VAWTs) exhibit enhanced resilience to elevated wind pressures linked to cyclones, necessitating reduced maintenance while providing dependable power generation in the cyclone-prone areas of Bangladesh. Implementing VAWTs in rural and coastal regions can establish a more reliable and robust energy supply, hence reducing interruptions caused by natural disasters. Additionally, VAWTs are effective because they do not require installation at heights of 100 feet or more, unlike HAWTs, making them more accessible and easier to maintain, especially in remote or challenging locations.

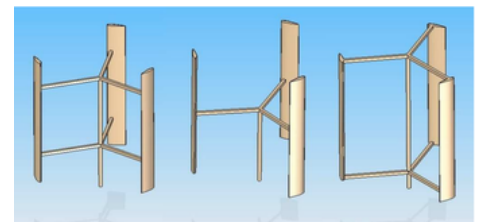


Fig: Rotor configurations of VAWTs (Castillo, 2011)

2. Hybrid Wind-Solar Systems with Offshore Deployment

Bangladesh's long coastline, which includes Mongla, Sandwip, and Saint Martin's Island, has steady wind speeds that are good for mixed offshore projects. Using both wind and solar power together in hybrid systems makes sure that there is a steady flow of electricity. Solar power can help with generation when there isn't much wind, making the system more stable and reliable. This mixed method keeps energy output stable, making sure it's always available, even when the weather or seasons change.

3. Advanced Energy Storage Solutions for Grid Resilience

Centralized energy storage facilities at government-operated substations are vital to support this hybrid model. Battery systems can retain excess energy produced during peak wind or solar periods, guaranteeing dependable energy availability during low-production intervals. These storage devices are vital for off-grid populations, allowing them to maintain connectivity during natural disasters. Centralized storage enhances system modernisation, guaranteeing that all communities, particularly those in remote or disaster-prone regions, have access to steady and reliable power (IEPMP, 2023).

Economic Solution

The economic viability of wind energy in Bangladesh depends on minimizing substantial early expenditures and investing in locally tailored, efficient technologies. Our economic approach prioritizes research and development, infrastructural collaboration, and governmental assistance.

1. Investment in Local Research and Development (R&D)

Investment in research and development targeting technology designed for Bangladesh's climate and geography can lower expenses and diminish dependence on imports. Investment in research and development for vertical axis wind turbines (VAWTs) can facilitate the creation of cyclone-resistant, cost-effective turbines, hence generating opportunities for local industry and reducing project costs (Shuvo, 2023). The Mongla project provides electricity at a competitive rate of Tk 12.5 per kWh, illustrating that localized solutions may be both efficient and cost-effective (The Daily Star, 2022).

2. Cost Sharing through Hybrid Infrastructure

Hybrid wind-solar systems enable the sharing of transmission lines, substations, and storage facilities, reducing capital and operational costs. Hybrid projects can reduce infrastructure expenses by consolidating resources, hence enhancing their economic viability and scalability for the spread of renewable energy in Bangladesh (Shuvo, 2023).

3. Government-Backed Centralized Storage Facilities

Government investment in centralized energy storage at substations can enhance the appeal of wind energy to commercial developers by lowering project costs, increasing system stability, and ensuring a stable electricity supply during peak demand periods. Centralized storage technologies not only reduce financial risks for investors but also guarantee reliable energy availability, especially during natural disasters, hence enhancing both economic stability and energy security (IEPMP, 2023).

Social and Environmental Solution

1. Expanding Energy Access for Remote and Off-Grid Areas

Offshore wind farms can provide dependable energy to isolated, off-grid settlements, especially in Saint Martin, Kuakata, and Sandwip. Enhancing energy access in these regions will improve living standards, generate local employment opportunities, and bolster local economies. Wind energy can bolster disaster resilience by providing a consistent power supply in remote places, hence maintaining community connectivity to the grid during extreme weather events (Uddin & Rahman, 2019).

2. Enhancing Energy Security and Reducing Dependence on Fossil Fuels

The increase of wind energy can diminish Bangladesh's reliance on imported fossil fuels, stabilise electricity prices, and mitigate susceptibility to global market volatility. An eclectic energy portfolio featuring a substantial renewable segment will enhance Bangladesh's energy security and alleviate the environmental repercussions of fossil fuels, thereby advancing the nation's international climate objectives and reducing its carbon footprint (Miskat et al., 2020).

3. Cyclone-Resilient and Environmentally Friendly Infrastructure

In parts of Bangladesh that are prone to natural disasters, it is essential to install infrastructure that is resistant to cyclones, such as VAWTs. These turbines are engineered to endure high winds and maintain operation during severe weather, guaranteeing consistent and stable energy output. Cyclone-resistant turbines mitigate the environmental and economic expenses linked to fossil fuel alternatives, which frequently experience interruptions during natural disasters (Jacobson et al., 2018).

4. Public Awareness and Community Involvement

Public acceptance and community support are essential for the success of wind energy projects. Campaigns that teach people about the economic, environmental, and social benefits of wind energy can help get people in the area to support it. Community involvement can help people feel better about wind energy and get more people involved in renewable energy projects (Chowdhury, 2020), especially in rural areas where people are worried about how the land is used and how it affects the environment. These efforts will show how reliable and strong modern wind technologies are. They will also show how hybrid models and grid modernisation offer consistent, community-focused solutions.

Expert Opinion

Renewable energy specialists underscore the necessity of customised solutions for Bangladesh's distinct environmental and geographic circumstances, particularly via hybrid energy models, centralised storage, and localised investment in wind technologies. Regional specialists recommend the implementation of hybrid wind-solar systems and Vertical Axis Wind Turbines (VAWTs) that are capable of withstanding Bangladesh's unpredictable wind patterns and severe weather conditions. By concentrating on regionally tailored solutions, Bangladesh may establish itself as a pioneer in renewable energy throughout South Asia.

Shayan Shafi, Energy Team Leader at the U.S. Agency for International Development (USAID), asserts that effective renewable energy projects necessitate a rigorous methodology for resource evaluation, site selection, and grid integration. Shafi underscores that the primary considerations are resource evaluation, appropriate site selection, equipment transfer, establishment of power evacuation facilities such as linking the project to grid substations, cost evaluation, and attaining an economic scale. He elucidates that these elements are the fundamental basis of any renewable energy initiative.

Shafi emphasises that the project's configuration—whether utilising VAWTs, offshore wind installations, or hybrid systems—must be customised to the particular site characteristics and opportunity for innovation. Critical aspects like tower height, turbine design, wind velocity, and direction must be meticulously assessed. Although offshore wind and hybrid systems present advantageous options, Shafi notes that they entail supplementary expenses, such as specialised maintenance, storage facilities, and improved service mechanisms. He recommends that "a benefit-cost analysis is crucial for enhancing production and sustainability in these projects." A comprehensive evaluation will guarantee the sustainable viability of renewable initiatives in Bangladesh.

Mr. Mehedi Islam Aneek, Assistant Vice President of the Project Management Department at Orion Power, offers additional insights into the economic aspect. Aneek suggests that government investment in centralised energy storage facilities will make renewable energy projects more economically viable for the private sector. Centralised storage may assist companies in addressing issues associated with the intermittent supply of renewable energy, hence enhancing the attractiveness and economic

viability of investments in wind and solar installations. Aneek assumes that, with government-supported storage, private enterprises might offer renewable energy to the government at lower rates, potentially as low as 3 to 4 BDT per unit. The reduced cost may promote increased adoption of renewable energy within the sector, expediting the transition to sustainable energy. Uddin and Rahman (2019) demonstrate the effectiveness of locally customised wind energy projects, indicating that customised wind systems may have significant output potential. Their findings suggest that innovative and resilient wind-solar hybrid solutions could not only meet local energy needs but also transform Bangladesh into a model for renewable energy in the region.

Conclusion

Wind energy offers a pivotal opportunity for Bangladesh to sustainably meet its energy requirements, lower reliance on fossil fuels, and enhance climate resilience amid escalating natural disasters. By harnessing the unexploited potential of Bangladesh's coastline through innovative Vertical Axis Wind Turbines (VAWTs), hybrid wind-solar systems, and efficient centralised storage solutions, Bangladesh can establish a dependable, resilient, and community-oriented renewable energy infrastructure. These customised solutions guarantee a reliable power supply, safeguard at-risk isolated communities during severe weather, and foster local economic development.

Strategic investments in research and development, collaborative infrastructure in hybrid initiatives, and improved public participation can render wind energy accessible and economical, benefiting all populations. Through joint assistance from the government, corporate sector, and local communities, Bangladesh can achieve its vision of a sustainable energy future, establishing a standard for renewable energy development in the region. Wind energy serves not merely as a technological remedy but as a conduit for enhanced energy security, environmental responsibility, and social progress, establishing the groundwork for a robust and thriving Bangladesh.

Call to Action

To fully harness its wind energy potential, the Bangladeshi government must prioritise strategic research and development investments, advocate for hybrid wind-solar infrastructure, and modernise grid systems to facilitate resilient energy storage. Public engagement and community involvement are essential for cultivating local support. Through these projects, Bangladesh may convert its coastal wind resources into a sustainable energy infrastructure, yielding enduring advantages for both the populace and the environment.

References

1. Babu, M. T., Nei, H., & Kowser, M. A. (2022). Prospects and necessity of wind energy in Bangladesh for the forthcoming future. *Journal of the Institute of Engineers India Series C*, 103(4), 913–929. <https://doi.org/10.1007/s40032-022-00834-8>
2. Chowdhury, S. (2020). National solar energy roadmap 2021-2041. Sustainable and Renewable Energy Development Authority (SREDA). <https://climateportal.ccdbbd.org/wp-content/uploads/2021/03/National-Solar-Energy-Roadmap.pdf>
3. Integrated Energy and Power Master Plan (IEPMP). (2023). Power Division, Government of Bangladesh. https://powerdivision.portal.gov.bd/sites/default/files/files/powerdivision.portal.gov.bd/page/4f81bf4d_1180_4c53_b27c_8fa0eb11e2c1/IEPMP%202023.pdf
4. Jacobson, M. Z., Delucchi, M. A., Cameron, M. A., & Frew, B. A. (2018). Low-cost solutions to the grid reliability problem with 100% penetration of intermittent wind, water, and solar for all purposes. *Proceedings of the National Academy of Sciences*, 112(49), 15060-15065. <https://doi.org/10.1073/pnas.1510028112>
5. Miskat, M. I., Ahmed, A., Rahman, M. S., Chowdhury, H., Chowdhury, T., Chowdhury, P., Sait, S. M., & Park, Y. K. (2020). An overview of the hydropower production potential in Bangladesh to meet the energy requirements. *Environmental Engineering Research*, 26(6), Article 200514. <https://doi.org/10.4491/eer.2020.514>
6. Shuvo, M. (2023, July 2). The wind energy potential of Cox's Bazar. *The Daily Star*. <https://bangla.thedailystar.net/environment/natural-resources/electricity-fuel/news-492941>
7. The Daily Star. (2022, September 4). Mongla wind power project: BPDB to buy electricity for 20 years. <https://www.thedailystar.net/business/news/mongla-wind-power-project-bpdb-buy-electricity-20-years-3123801>
8. Uddin, M. N., & Rahman, M. A. (2019). Feasibility study of wind power generation at different sites in Bangladesh. *Energy Procedia*, 160, 655-661. <https://doi.org/10.1016/j.egypro.2019.02.218>
9. Castillo, J. (2011). *Small-scale vertical axis wind turbine design* (Bachelor's thesis, Universitat Politècnica de Catalunya). UPCommons. <https://upcommons.upc.edu/bitstream/handle/2099.1/19136/memoria.pdf>